Projectile Motion

Use alongside the Projectile Motion slides.

NAME	CLASS	DATE

Checkpoints
Each checkpoint matches a group of slides. Complete it when your teacher pauses there.

Constants
Use $g = 10\ \mathrm{m\,s^{-2}}$, as the slides do. Ignore air resistance unless told otherwise.

Signs & working
Up and right are positive. Show your working — a correct method earns marks even if the number slips.

01 The big ideas

SLIDES 1–4

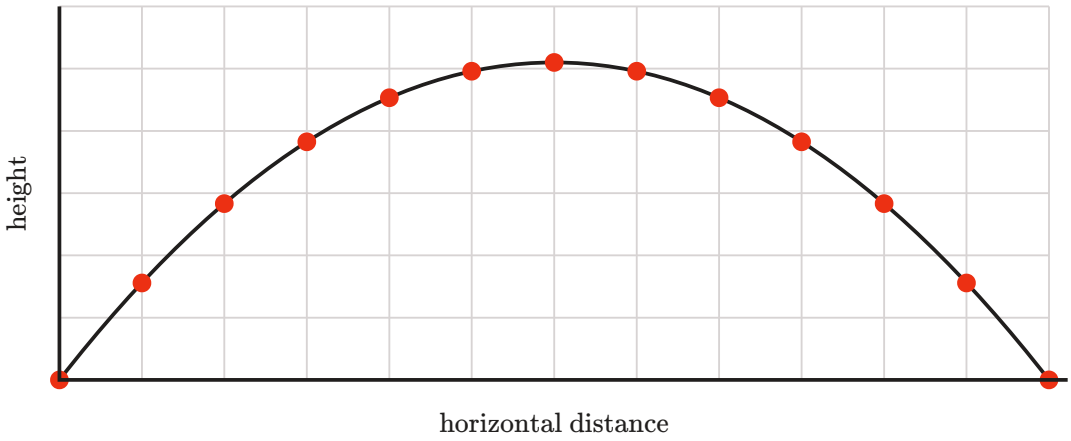
1.1 Complete the statements for a projectile with no air resistance.

Horizontal acceleration: $a_x =$ _____ Vertical acceleration: $a_y =$ _____

1.2 On slide 2, the orange dots along the bottom are equally spaced but the teal dots up the side are not. What does each set of dots tell you about the motion?

1.3 Ball P is dropped from a table. At the same moment, ball Q is pushed off the same table horizontally at $3\ \mathrm{m\,s^{-1}}$. Which ball lands first? Explain using the word *independent*.

1.4 The dots show a ball with no air resistance. Sketch the path the same ball would follow if air resistance were **not** negligible (slide 4). Then complete the sentences.



With air resistance the peak is _____ and to the _____.

The horizontal distance before the peak is _____ than the distance after the peak.

02 Skills lab: resolving vectors

BEFORE SLIDE 8

Almost every projectile question starts by splitting the launch velocity into a horizontal part and a vertical part. If this step goes wrong, everything after it goes wrong too — so we practise it properly here.

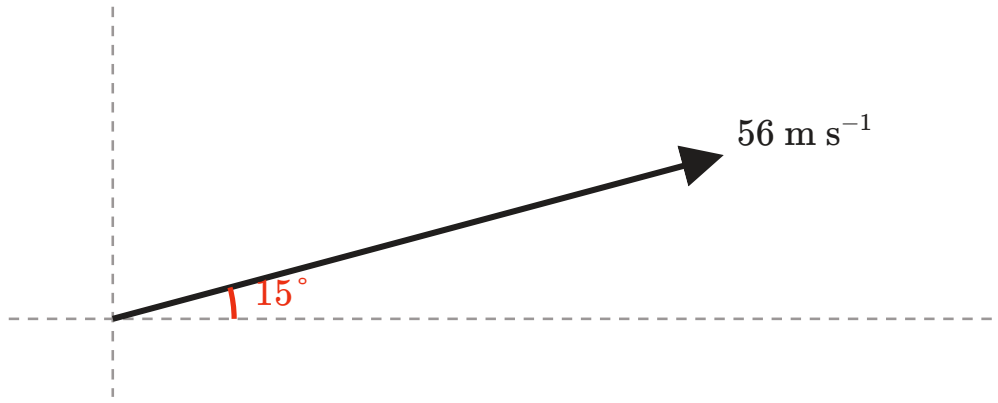
THE METHOD THAT ALWAYS WORKS

1. Draw the vector and mark the angle you are given.
2. Draw the two components as dashed arrows, making a right-angled triangle with the vector as the hypotenuse.
3. The component **next to** the angle (adjacent) $= v\cos\theta$.
4. The component **opposite** the angle $= v\sin\theta$.
5. Give each component a sign: $+$ for right and up, $-$ for left and down.
6. Check: $\sqrt{u_x^2 + u_y^2}$ should give back the original magnitude.

Watch out: "x is always $\cos$" is only true when the angle is measured from the horizontal. Always look at where the angle is.

2.1 **Calculator check.** Work out $\sin 30^\circ =$ _____. If you did not get 0.5, your calculator is in radians. Switch it to degrees now.

2.2 **Guided example** (the cannon on slide 8). A cannon fires a ball at $56\ \mathrm{m\,s^{-1}}$ at 15° above the horizontal.



Step 1 Draw the horizontal and vertical components on the diagram as dashed arrows.

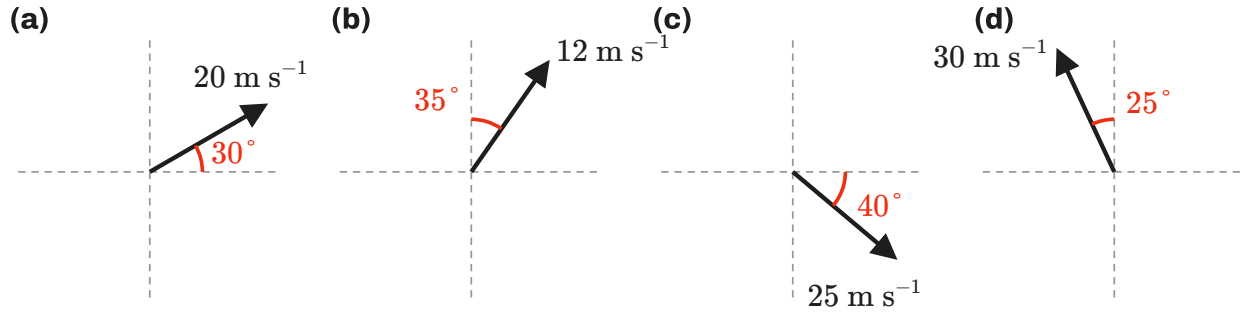
Step 2 Which component is adjacent to the 15° angle? _____

Step 3 Before calculating, predict which component is bigger and say why.

Step 4 $u_x = 56 \times$ _____ $15^\circ =$ _____ $\mathrm{m\,s^{-1}}$ $\quad$ $u_y = 56 \times$ _____ $15^\circ =$ _____ $\mathrm{m\,s^{-1}}$

Step 5 Check: $\sqrt{u_x^2 + u_y^2} =$ _____. Does this give back $56\ \mathrm{m\,s^{-1}}$? _____

2.3 **Practice.** Draw the components on each diagram first. Look carefully at where each angle is measured from.



	Angle measured from horizontal / vertical	**Adjacent component** x or y	u_x / $\mathrm{m\,s^{-1}}$ with sign	u_y / $\mathrm{m\,s^{-1}}$ with sign	**Check** $\sqrt{u_x^2 + u_y^2}$
(a)					
(b)					
(c)					
(d)					

2.4 Spot the mistake. Each student made one error. Say what went wrong and give the correct answer.

Student working	What went wrong, and the correct answer
Cannon at 15° $u_x = 56 \sin 15^\circ = 14.5\ \mathrm{m\,s^{-1}}$	
Vector (a) $u_x = 20 \cos 30^\circ = 3.09\ \mathrm{m\,s^{-1}}$	
Vector (c) $u_y = 25 \sin 40^\circ = +16.1\ \mathrm{m\,s^{-1}}$	
Vector (b) $u_x = 12 \cos 35^\circ = 9.83\ \mathrm{m\,s^{-1}}$	

2.5 Going backwards: components to a vector. You will need this on slides 17 and 25.

COMBINING COMPONENTS

Magnitude

$$v = \sqrt{v_x^2 + v_y^2}$$

Angle to the horizontal

$$\tan\theta = \frac{|v_y|}{|v_x|}$$

Opposite over adjacent, with the angle at the horizontal. Always say whether the angle is **above** or **below** the horizontal.

(a) A stone has $v_x = 18\ \mathrm{m\,s^{-1}}$ and $v_y = -26\ \mathrm{m\,s^{-1}}$. Sketch the velocity triangle, then find the speed and the angle the velocity makes with the horizontal.

WORKING

(b) A ball has $v_x = 8.0\ \mathrm{m\,s^{-1}}$ and $v_y = +25\ \mathrm{m\,s^{-1}}$. Find its speed and direction.

WORKING

03 From 1D to 2D equations

SLIDES 5–7

3.1 Start from the 2D kinematic equations on slide 5. Substitute your accelerations from 1.1 to get the reduced equations of projectile motion (slide 6).

	Horizontal (x)	**Vertical (y)**
Acceleration		
Displacement equation after substituting		
Velocity equation after substituting		

3.2 Slide 7: show that the path of a projectile is a parabola.

Step 1 Rearrange $\Delta x = u_x t$ to make t the subject: $t =$ _____

Step 2 Substitute this into $\Delta y = u_y t - 5t^2$: $\Delta y =$ _____

Step 3 Compare with $y = Ax + Bx^2$: $A =$ _____ $B =$ _____

Step 4 Why must B be negative? What would the path look like if it were positive?

04 The cannon

SLIDES 8–11

Use your own values of u_x and u_y from 2.2, to 3 significant figures.

4.1 Write the tailored equations for this cannon ball (slide 9).

$\Delta x =$ _____ $v_x =$ _____

$\Delta y =$ _____ $v_y =$ _____

4.2 At maximum height, $v_y =$ _____. Use this to find the time taken to reach maximum height.

WORKING

4.3 **EXTENSION** Find the maximum height reached.

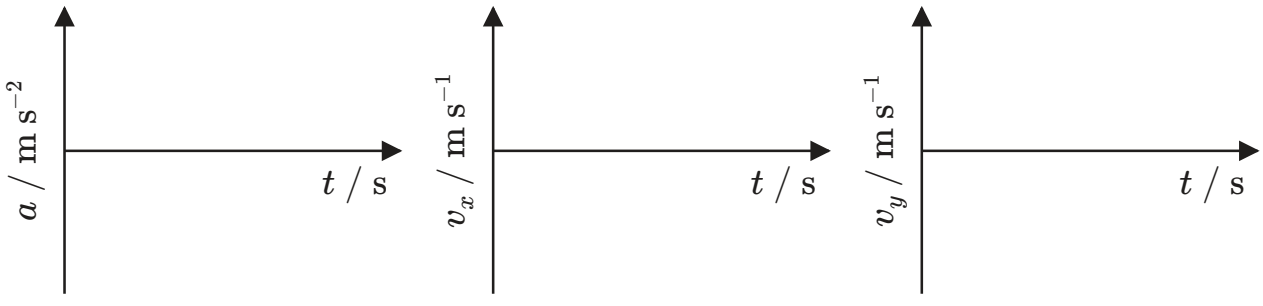
WORKING

4.4 When the ball returns to the height of the muzzle, $\Delta y =$ _____. Find the time of flight and the range (slide 10).

WORKING

4.5 The slides round u_y to $15\ \mathrm{m\,s^{-1}}$ and get a range of $162\ \mathrm{m}$. Compare this with your answer. What does this tell you about rounding in the middle of a calculation?

4.6 Sketch the three graphs for the flight (slide 11). Label the starting value, the final value, and the time at maximum height.



05 Launched from a cliff

SLIDES 12–18

5.1 Slide 12. At maximum height, $v_y =$ _____ and the acceleration is _____. My answer: _____

Why is the acceleration **not** zero at the top of the path?

5.2 Slide 13. My answer: _____. Complete: the time of flight depends only on the _____ motion, so u_x can only affect the _____.

5.3 Slides 14–17: the stone thrown horizontally at $18\ \mathrm{m\,s^{-1}}$ from a $33\ \mathrm{m}$ cliff.

(a) Fill in what you know first.

u_x	u_y	a_x	a_y	Δy check the sign

(b) Find the time taken to reach sea level.

WORKING

(c) Find the distance of the stone from the base of the cliff when it reaches sea level.

WORKING

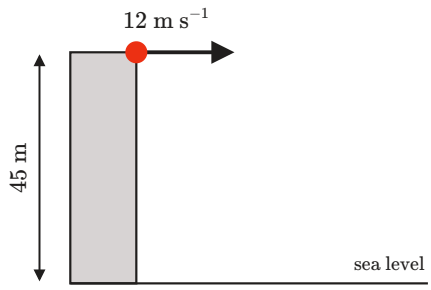
(d) At sea level, $v_x =$ _____ and $v_y =$ _____. Sketch the velocity triangle (just like 2.5) and find the angle the velocity makes with the sea.

WORKING

5.4 Slide 18. My answer: _____. One-sentence reason:

5.5 **Your turn.** A stone is thrown horizontally at $12\ \mathrm{m\,s^{-1}}$ from a cliff $45\ \mathrm{m}$ high.

Find **(a)** the time to reach sea level, **(b)** the horizontal distance travelled, and **(c)** the speed and direction of the stone as it hits the sea.



WORKING

06 Using energy instead

SLIDES 19–21

6.1 Slide 19: a $0.44\ \mathrm{kg}$ stone thrown horizontally at $22\ \mathrm{m\,s^{-1}}$ from a $32\ \mathrm{m}$ cliff.

Kinetic energy at the top, $\frac{1}{2}mu^2$	_____ J
Gravitational potential energy lost on the way down, mgh	_____ J
Total kinetic energy at sea level	_____ J

6.2 Slide 20: 34% of the total energy is transferred due to air resistance. What percentage is left as kinetic energy? _____ %. Use this to find the speed at sea level.

WORKING

6.3 Slide 21. My answer: _____. Explain why h does not appear in the correct expression.

6.4 The energy method found the final speed without resolving any vectors. Why can it **not** tell you the direction the stone is moving?

07 Clearing the wall

SLIDES 22–23

7.1 Resolve $28\ \mathrm{m\,s^{-1}}$ at 30° above the horizontal. Use the skills lab steps: sketch first.

$u_x =$ _____ $\mathrm{m\,s^{-1}}$ $\quad$ $u_y =$ _____ $\mathrm{m\,s^{-1}}$

7.2 Which motion, horizontal or vertical, tells you **when** the ball reaches the wall? _____ Find that time.

WORKING

7.3 Use that time in the vertical equation to find the maximum height of the wall.

WORKING

08 Reading a trajectory graph

SLIDES 24–27

8.1 The dots are $0.50\ \text{s}$ apart. Horizontal distance between dots = _____ m, so u_x = _____ m s^{-1}.

8.2 Use the peak. Horizontal distance to the peak = _____ m, so time to the peak = _____ s. At the peak $v_y = 0$, so u_y = _____ m s^{-1}.

8.3 After $3.0\ \text{s}$, read Δx = _____ m and Δy = _____ m from the graph. Combine them (this is 2.5 again) to find the magnitude and direction of the displacement.

WORKING

8.4 Slide 27: describe two ways the path changes if air resistance is not negligible.

Exit check: how confident am I?

I CAN…	NOT YET	NEARLY	CONFIDENT
resolve a velocity when the angle is measured from the **horizontal**	☐	☐	☐
resolve a velocity when the angle is measured from the **vertical**	☐	☐	☐
give each component the correct sign	☐	☐	☐
combine two components into a speed and a direction	☐	☐	☐
explain why horizontal and vertical motion are independent	☐	☐	☐
decide when to use the equations of motion and when to use energy	☐	☐	☐